

✓ Steam Game Success Prediction

MSc Information Technology Management — Machine Learning and Visualization for Data

Author: Ibrahim Eren **Dataset:** Steam Games Dataset (Kaggle — fronkongames), 122,611 games **Objective:** Predict whether a game will be successful based on its features, and identify game archetypes through clustering.

✓ Step 1: Loading the Data

In this step we mount Google Drive and load the Steam games dataset into a pandas DataFrame.

```
# =====  
# STEP 1: Loading the Data  
# =====  
import os  
import pandas as pd  
import gdown  
  
# Shared dataset (Google Drive – anyone with the link can view)  
DRIVE_LINK = 'https://drive.google.com/file/d/1fR51Z0nH3LlRBFk0bbKm0aPU9e4UiPDW/view?usp=sharing'  
OUTPUT      = 'games.csv'  
  
print("Downloading the dataset from Google Drive...")  
# fuzzy=True lets gdown parse the full share link and handle the  
# large-file virus-scan warning page automatically.  
gdown.download(url=DRIVE_LINK, output=OUTPUT, quiet=False, fuzzy=True)  
  
# Sanity check: real file is ~250 MB; anything tiny means it downloaded  
# Google Drive's HTML warning page instead of the CSV.  
size_mb = os.path.getsize(OUTPUT) / 1024 / 1024  
print(f"Downloaded file size: {size_mb:.1f} MB")  
assert size_mb > 50, "Download too small – likely an HTML error page. Check Drive sharing permissions."  
  
# Read the CSV  
df = pd.read_csv(OUTPUT, low_memory=False)  
print(f"Dataset loaded successfully! Shape: {df.shape}")  
df.head()
```

Downloading the dataset from Google Drive...

Downloading...

From (original): <https://drive.google.com/uc?id=1fR51Z0nH3LLRBFk0bbKm0aPU9e4UiPDW>

From (redirected): <https://drive.google.com/uc?id=1fR51Z0nH3LLRBFk0bbKm0aPU9e4UiPDW&confirm=t&uuid=b9318>

To: /content/games.csv

100%|██████████| 389M/389M [00:07<00:00, 52.7MB/s]

Downloaded file size: 371.1 MB

Dataset loaded successfully! Shape: (122611, 39)

	AppID	Name	Release date	Estimated owners	Peak CCU	Required age	Price	Discount	DLC count	About the game	Supported Languages	...	p1
2539430	Black Dragon Mage Playtest	Aug 1, 2023	0 - 0	0	0	0.00	0		0	NaN	[]	...	
496350	Supipara - Chapter 1 Spring Has Come!	Jul 29, 2016	0 - 20000	0	0	5.24	65		0	Springtime, April: when the cherry trees come ...	['English']	...	
1034400	Mystery Solitaire The Black Raven	May 6, 2019	0 - 20000	0	0	4.99	0		0	Immerse yourself in the most beloved, mystical...	['English', 'French', 'German', 'Russian']	...	
3292190	버튜버 파 라노이아 - Vtuber Paranoia	Oct 31, 2024	0 - 20000	1	0	8.99	0		1	synopsis 'Hello, I'm Hiyoro, a new YouTuber!' ...	['Korean']	...	
3631080	Maze Quest VR	Apr 24, 2025	0 - 20000	0	0	4.99	0		0	Its not just a Maze; its a Quest! Enter the ca...	['English']	...	

5 rows x 39 columns

✓ Step 2: First Look at the Data (Exploratory Data Analysis)

Here we examine the dataset structure, check for missing values, and explore the key columns we will use to define game success.

```
# =====
# STEP 2: First Look at the Data (EDA)
# =====

# 2.1 General information about columns and data types
print("==== COLUMN INFO =====")
df.info()

# 2.2 Missing values in the most important columns
print("\n==== MISSING VALUES (key columns) =====")
key_cols = ['Name', 'Price', 'Positive', 'Negative', 'Genres', 'Release date', 'Estimated owners']
for col in key_cols:
    missing = df[col].isna().sum()
    print(f"{col}: {missing}, missing ({missing/len(df)*100:.1f}%)")

# 2.3 Basic statistics for the review columns (the basis of "success")
print("\n==== REVIEW STATISTICS =====")
print(df[['Positive', 'Negative']].describe())
```

```

10 Full audio languages      122611 non-null object
11 Reviews                   12070 non-null object
12 Header image             122530 non-null object
13 Website                   49676 non-null object
14 Support url               54142 non-null object
15 Support email             100348 non-null object
16 Windows                   122611 non-null bool
17 Mac                       122611 non-null bool
18 Linux                     122611 non-null bool
19 Metacritic score          122611 non-null int64
20 Metacritic url            4256 non-null object
21 User score                122611 non-null int64
22 Positive                  122611 non-null int64
23 Negative                  122611 non-null int64
24 Score rank                40 non-null float64
25 Achievements              122611 non-null int64
26 Recommendations          122611 non-null int64
27 Notes                     22458 non-null object
28 Average playtime forever  122611 non-null int64
29 Average playtime two weeks 122611 non-null int64
30 Median playtime forever   122611 non-null int64
31 Median playtime two weeks 122611 non-null int64
32 Developers                114174 non-null object
33 Publishers                 113702 non-null object
34 Categories                 113658 non-null object
35 Genres                     114198 non-null object
36 Tags                       83346 non-null object
37 Screenshots                116593 non-null object
38 Movies                     0 non-null float64
dtypes: bool(3), float64(3), int64(14), object(19)
memory usage: 35.0+ MB

```

```

===== MISSING VALUES (key columns) =====
Name: 0 missing (0.0%)
Price: 0 missing (0.0%)
Positive: 0 missing (0.0%)
Negative: 0 missing (0.0%)
Genres: 8,413 missing (6.9%)
Release date: 0 missing (0.0%)
Estimated owners: 0 missing (0.0%)

```

```

===== REVIEW STATISTICS =====
          Positive      Negative
count  1.226110e+05  1.226110e+05
mean    1.044986e+03  1.691974e+02
std      2.809173e+04  5.374645e+03
min       0.000000e+00  0.000000e+00
25%       0.000000e+00  0.000000e+00
50%       5.000000e+00  1.000000e+00
75%       3.700000e+01  1.000000e+01
max       7.642084e+06  1.173003e+06

```

Step 3: Defining Success with the Wilson Score

A simple positive ratio is misleading because a game with 5 positive reviews would score 100%, same as a game with millions of reviews. To balance review quality and review volume, we use the Wilson Lower Bound score, which gives a statistically reliable success measure.

```

# =====
# STEP 3: Defining Success with the Wilson Score
# =====
import numpy as np
from scipy import stats

# 3.1 Total reviews and simple positive ratio
df['total_reviews'] = df['Positive'] + df['Negative']
df['positive_ratio'] = df['Positive'] / df['total_reviews'].replace(0, np.nan)

# 3.2 Wilson Lower Bound score
# This balances the ratio with the number of reviews (confidence).
# Few reviews -> lower score (uncertain). Many reviews -> closer to true ratio.
def wilson_lower_bound(positive, total, confidence=0.95):
    if total == 0:
        return 0
    z = stats.norm.ppf(1 - (1 - confidence) / 2)
    phat = positive / total
    denominator = 1 + z**2 / total
    centre = phat + z**2 / (2 * total)
    margin = z * np.sqrt((phat * (1 - phat) + z**2 / (4 * total)) / total)

```

```

        return (centre - margin) / denominator

# Apply the Wilson score to every game
df['wilson_score'] = df.apply(
    lambda row: wilson_lower_bound(row['Positive'], row['total_reviews']),
    axis=1
)

# 3.3 Compare three example games to see how Wilson works
print("==== EXAMPLE COMPARISON =====")
examples = pd.DataFrame({
    'Positive': [5, 1000, 100000],
    'Negative': [0, 100, 15000],
})
examples['total'] = examples['Positive'] + examples['Negative']
examples['ratio'] = examples['Positive'] / examples['total']
examples['wilson'] = examples.apply(
    lambda r: wilson_lower_bound(r['Positive'], r['total']), axis=1
)
print(examples)

```

```

==== EXAMPLE COMPARISON =====
   Positive  Negative  total  ratio  wilson
0         5         0      5  1.000000  0.565518
1      1000        100     1100  0.909091  0.890649
2     100000     15000    115000  0.869565  0.867606

```

✓ Step 4: Setting the Success Threshold

Now we examine the distribution of Wilson scores across all games and define a data-driven threshold to label each game as successful (1) or not successful (0).

```

# =====
# STEP 4: Setting the Success Threshold
# =====

# 4.1 Look only at games that have at least some reviews
# (games with 0 reviews cannot have a meaningful success score)
games_with_reviews = df[df['total_reviews'] > 0].copy()
print(f"Games with at least 1 review: {len(games_with_reviews):,}")
print(f"Games with at least 10 reviews: {(df['total_reviews'] >= 10).sum():,}")
print(f"Games with at least 50 reviews: {(df['total_reviews'] >= 50).sum():,}")

# 4.2 Distribution of Wilson scores (for games with >= 10 reviews)
print("\n==== WILSON SCORE DISTRIBUTION (>= 10 reviews) =====")
subset = df[df['total_reviews'] >= 10]
print(subset['wilson_score'].describe())

# 4.3 Show percentiles to help choose a threshold
print("\n==== PERCENTILES =====")
for p in [50, 60, 70, 75, 80, 90]:
    val = subset['wilson_score'].quantile(p/100)
    print(f"{p}th percentile: {val:.3f}")

```

```

Games with at least 1 review: 82,949
Games with at least 10 reviews: 56,655
Games with at least 50 reviews: 30,620

```

```

==== WILSON SCORE DISTRIBUTION (>= 10 reviews) =====
count    5.665500e+04
mean      6.515617e-01
std       2.055254e-01
min      -1.173174e-17
25%       5.212421e-01
50%       6.873739e-01
75%       8.128800e-01
max       9.945528e-01
Name: wilson_score, dtype: float64

```

```

==== PERCENTILES =====
50th percentile: 0.687
60th percentile: 0.741
70th percentile: 0.788
75th percentile: 0.813
80th percentile: 0.838
90th percentile: 0.892

```

✓ Step 5: Creating the Target Variable

We define a game as successful (1) if it has at least 10 reviews AND a Wilson score of 0.80 or higher, aligning with Steam's "Very Positive" rating standard. Otherwise it is labelled not successful (0).

```
# =====
# STEP 5: Creating the Target Variable
# =====

# 5.1 Keep only games with enough reviews to be meaningful
# (at least 10 reviews -> avoids noise from games with 1-2 reviews)
df_model = df[df['total_reviews'] >= 10].copy()
print(f"Games used for modelling: {len(df_model):,}")

# 5.2 Create the binary target: success = 1, not success = 0
# Threshold: Wilson score >= 0.80 (aligned with Steam 'Very Positive')
SUCCESS_THRESHOLD = 0.80
df_model['success'] = (df_model['wilson_score'] >= SUCCESS_THRESHOLD).astype(int)

# 5.3 Check the class balance
print("\n===== TARGET DISTRIBUTION =====")
print(df_model['success'].value_counts())
print("\nPercentages:")
print(df_model['success'].value_counts(normalize=True).round(3) * 100)
```

Games used for modelling: 56,655

```
===== TARGET DISTRIBUTION =====
success
0    41085
1    15570
Name: count, dtype: int64
```

```
Percentages:
success
0     72.5
1     27.5
Name: proportion, dtype: float64
```

✓ Step 6: Feature Engineering

We prepare the input features for the model. This includes cleaning the 'Estimated owners' range column, extracting the release year, creating price bands, and counting the number of genres, categories, and platforms each game supports.

```
# =====
# STEP 6 (FINAL FIX): Use the correctly-aligned columns
# =====
# The CSV columns are shifted by one position. The labels are wrong,
# but the DATA is intact - we just read each value from the correct label.

# REAL release date is stored under the 'Name' label -> extract year
df_model['release_year'] = df_model['Name'].str.extract(r'(\d{4})').astype(float)

# REAL estimated owners is stored under the 'Release date' label (e.g. "0 - 20000")
def parse_owners(value):
    try:
        low, high = str(value).split(' - ')
        return (int(low) + int(high)) / 2
    except:
        return 0

df_model['owners_avg'] = df_model['Release date'].apply(parse_owners)

# REAL price is stored under the 'Required age' label
df_model['price_real'] = pd.to_numeric(df_model['Required age'], errors='coerce')

# --- Check the results ---
print("===== release_year (fixed) =====")
print(df_model['release_year'].describe())

print("\n===== owners_avg (fixed) =====")
print(df_model['owners_avg'].describe())
print("Non-zero count:", (df_model['owners_avg'] > 0).sum())
```

```
print("\n==== price_real (fixed) =====")
print(df_model['price_real'].describe())
```

```
==== release_year (fixed) =====
count      56655.000000
mean       2019.885818
std         3.491388
min        1997.000000
25%        2017.000000
50%        2020.000000
75%        2023.000000
max        2025.000000
Name: release_year, dtype: float64
```

```
==== owners_avg (fixed) =====
count      5.665500e+04
mean       1.748231e+05
std        1.893175e+06
min        1.000000e+04
25%        1.000000e+04
50%        1.000000e+04
75%        3.500000e+04
max        1.500000e+08
Name: owners_avg, dtype: float64
Non-zero count: 56655
```

```
==== price_real (fixed) =====
count      56655.000000
mean        5.211367
std         10.670322
min         0.000000
25%         0.990000
50%         2.990000
75%         6.290000
max        299.990000
Name: price_real, dtype: float64
```

```
# =====
# DIAGNOSIS: Check the raw values of two columns
# =====

# Check the raw 'Estimated owners' - is it text or number?
print("==== Estimated owners - data type and raw values =====")
print("Dtype:", df_model['Estimated owners'].dtype)
print(df_model['Estimated owners'].head(10).tolist())

print("\n==== Release date - raw values =====")
print("Dtype:", df_model['Release date'].dtype)
print(df_model['Release date'].head(10).tolist())
```

```
==== Estimated owners - data type and raw values =====
Dtype: int64
[0, 0, 8, 0, 1, 0, 13, 0, 0, 0]
```

```
==== Release date - raw values =====
Dtype: object
['0 - 20000', '0 - 20000', '0 - 20000', '0 - 20000', '0 - 20000', '0 - 20000', '100000 - 200000', '0 - 2
```

```
# =====
# DIAGNOSIS 2: See a full row to understand column shifting
# =====

# Print the first row with ALL columns and their values
print("==== FIRST ROW - every column =====")
first_row = df_model.iloc[0]
for col in df_model.columns:
    print(f"{col:30s} -> {str(first_row[col])[:50]}")
```

```
==== FIRST ROW - every column =====
AppID              -> Supipara - Chapter 1 Spring Has Come!
Name               -> Jul 29, 2016
Release date       -> 0 - 20000
Estimated owners   -> 0
Peak CCU           -> 0
Required age       -> 5.24
Price              -> 65
DiscountDLC count  -> 0
About the game     -> Springtime, April: when the cherry trees come into
```

```

Supported languages      -> ['English']
Full audio languages     -> []
Reviews                 -> nan
Header image            -> https://shared.akamai.steamstatic.com/store\_item\_a
Website                 -> http://mangagamer.org/supipara
Support url             -> http://mangagamer.com
Support email           -> support@mangagamer.com
Windows                 -> True
Mac                     -> False
Linux                   -> False
Metacritic score        -> 0
Metacritic url          -> nan
User score              -> 0
Positive                -> 252
Negative                -> 3
Score rank              -> nan
Achievements            -> 0
Recommendations         -> 231
Notes                   -> nan
Average playtime forever -> 8
Average playtime two weeks -> 0
Median playtime forever -> 8
Median playtime two weeks -> 0
Developers              -> minori
Publishers              -> MangaGamer
Categories               -> Single-player, Steam Trading Cards, Steam Cloud, Fami
Genres                  -> Adventure
Tags                    -> Adventure, Visual Novel, Anime, Cute
Screenshots             -> https://shared.akamai.steamstatic.com/store\_item\_a
Movies                  -> nan
total_reviews           -> 255
positive_ratio          -> 0.9882352941176471
wilson_score            -> 0.9659877937861515
success                 -> 1
release_year            -> 2016.0
owners_avg              -> 10000.0
price_real              -> 5.24

```

▼ Step 7: Preparing Features for Modelling

We select the final set of features for the model, handle the genre and category counts, and split the data into training and testing sets. The target variable is 'success' (1 = successful, 0 = not successful).

```

from sklearn.model_selection import train_test_split

# 7.1 Fix genre_count and category_count using the CORRECT columns
# (these were also affected, so we recount from the right labels)
df_model['genre_count'] = df_model['Genres'].fillna('').apply(
    lambda x: len(str(x).split(',')) if x else 0
)
df_model['category_count'] = df_model['Categories'].fillna('').apply(
    lambda x: len(str(x).split(',')) if x else 0
)

# 7.2 is_free based on the real price
df_model['is_free'] = (df_model['price_real'] == 0).astype(int)

# 7.3 Create platform_count
df_model['platform_count'] = df_model[['Windows', 'Mac', 'Linux']].sum(axis=1)

# 7.4 Select the final features for the model
feature_cols = [
    'price_real',      # game price
    'owners_avg',      # estimated owners (popularity proxy)
    'release_year',    # release year
    'platform_count',  # number of supported platforms
    'genre_count',     # number of genres
    'category_count',  # number of categories
    'is_free',         # is the game free
]

# 7.5 Build X (features) and y (target)
X = df_model[feature_cols].copy()
y = df_model['success']

# Handle any remaining missing values (fill with 0 just in case)
X = X.fillna(0)

```

```
# 7.6 Split into training (80%) and testing (20%) sets
# stratify=y -> keeps the same success ratio in both sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

print("==== DATA SPLIT ====")
print(f"Training set: {X_train.shape[0]:,} games")
print(f"Testing set: {X_test.shape[0]:,} games")
print(f"\nFeatures used: {feature_cols}")
print(f"\nSuccess ratio in training: {y_train.mean():.3f}")
print(f"Success ratio in testing: {y_test.mean():.3f}")
```

```
==== DATA SPLIT ====
Training set: 45,324 games
Testing set: 11,331 games

Features used: ['price_real', 'owners_avg', 'release_year', 'platform_count', 'genre_count', 'category_c

Success ratio in training: 0.275
Success ratio in testing: 0.275
```

▼ Step 8: Training the First Models

We train two classification models to predict game success: Logistic Regression (a simple linear baseline) and Random Forest (a more powerful ensemble model). Because the classes are imbalanced (27.5% successful), we use `class_weight='balanced'`. Features are scaled for Logistic Regression.

```
# =====
# STEP 8: Training the First Models
# =====

from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, roc_auc_score

# 8.1 Scale features (important for Logistic Regression)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# --- MODEL 1: Logistic Regression ---
print("==== MODEL 1: LOGISTIC REGRESSION ====")
logreg = LogisticRegression(class_weight='balanced', max_iter=1000, random_state=42)
logreg.fit(X_train_scaled, y_train)
y_pred_lr = logreg.predict(X_test_scaled)
y_proba_lr = logreg.predict_proba(X_test_scaled)[:, 1]

print(f"Accuracy: {accuracy_score(y_test, y_pred_lr):.3f}")
print(f"Precision: {precision_score(y_test, y_pred_lr):.3f}")
print(f"Recall: {recall_score(y_test, y_pred_lr):.3f}")
print(f"F1-Score: {f1_score(y_test, y_pred_lr):.3f}")
print(f"ROC-AUC: {roc_auc_score(y_test, y_proba_lr):.3f}")

# --- MODEL 2: Random Forest ---
print("\n==== MODEL 2: RANDOM FOREST ====")
rf = RandomForestClassifier(n_estimators=100, class_weight='balanced', random_state=42, n_jobs=-1)
rf.fit(X_train, y_train) # Random Forest does not need scaling
y_pred_rf = rf.predict(X_test)
y_proba_rf = rf.predict_proba(X_test)[:, 1]

print(f"Accuracy: {accuracy_score(y_test, y_pred_rf):.3f}")
print(f"Precision: {precision_score(y_test, y_pred_rf):.3f}")
print(f"Recall: {recall_score(y_test, y_pred_rf):.3f}")
print(f"F1-Score: {f1_score(y_test, y_pred_rf):.3f}")
print(f"ROC-AUC: {roc_auc_score(y_test, y_proba_rf):.3f}")
```

```
==== MODEL 1: LOGISTIC REGRESSION ====
Accuracy: 0.636
Precision: 0.389
Recall: 0.566
F1-Score: 0.461
ROC-AUC: 0.662
```

```
===== MODEL 2: RANDOM FOREST =====
Accuracy: 0.696
Precision: 0.440
Recall: 0.385
F1-Score: 0.410
ROC-AUC: 0.670
```

▼ Step 9: Adding Genre Features and Game Archetype Clusters

The first models used only numeric features and performed moderately. To improve them, we add the most common genres as binary features, then apply K-Means clustering to group games into archetypes (e.g. "premium AAA", "cheap indie"). The cluster label becomes a new feature that feeds the classification model.

```
# =====
# STEP 9.5: Build log features and genre one-hots (needed by Step 10)
# =====
import numpy as np

# Log-transformed price and owners (Step 10 expects these)
df_model['log_price'] = np.log1p(df_model['price_real'].fillna(0))
df_model['log_owners'] = np.log1p(df_model['owners_avg'].fillna(0))

# One-hot encode the top 10 most common genres (Step 10 expects genre_feature_cols)
genre_series = df_model['Genres'].fillna('').str.split(',')
all_genres = genre_series.explode().str.strip()
top10_genres = all_genres[all_genres != ''].value_counts().head(10).index.tolist()
print("Top 10 genres:", top10_genres)

for g in top10_genres:
    col = f'genre_{g.replace(" ", "_")}'
    df_model[col] = genre_series.apply(lambda lst: int(g in [x.strip() for x in lst]))

genre_feature_cols = [f'genre_{g.replace(" ", "_")}' for g in top10_genres]
print("genre_feature_cols ready:", genre_feature_cols)
print("log_price / log_owners ready:", 'log_price' in df_model.columns, 'log_owners' in df_model.columns)

Top 10 genres: ['Indie', 'Adventure', 'Action', 'Casual', 'Simulation', 'Strategy', 'RPG', 'Early Access']
genre_feature_cols ready: ['genre_Indie', 'genre_Adventure', 'genre_Action', 'genre_Casual', 'genre_Simulation', 'genre_Strategy', 'genre_RPG', 'genre_Early Access']
log_price / log_owners ready: True True
```

▼ Step 10: Final Model with All Features

We combine numeric features, genre features, and the K-Means archetype into one feature set, then retrain Logistic Regression and Random Forest. We expect the genre and archetype information to improve performance compared to the first models.

```
# =====
# STEP 10: Final Model with All Features
# =====

# 10.1 One-hot encode the archetype (turn it into 4 binary columns)
archetype_dummies = pd.get_dummies(df_model['archetype'], prefix='archetype')

# 10.2 Build the full feature set: numeric + genres + archetype
numeric_features = ['log_price', 'log_owners', 'release_year',
                    'platform_count', 'genre_count', 'category_count', 'is_free']

X_full = pd.concat([
    df_model[numeric_features].reset_index(drop=True),
    df_model[genre_feature_cols].reset_index(drop=True),
    archetype_dummies.reset_index(drop=True)
], axis=1).fillna(0)

y_full = df_model['success'].reset_index(drop=True)

# 10.3 Train-test split
X_train2, X_test2, y_train2, y_test2 = train_test_split(
    X_full, y_full, test_size=0.2, random_state=42, stratify=y_full
)
```

```
# 10.4 Scale for Logistic Regression
scaler2 = StandardScaler()
X_train2_scaled = scaler2.fit_transform(X_train2)
X_test2_scaled = scaler2.transform(X_test2)

# --- Logistic Regression ---
print("==== LOGISTIC REGRESSION (full features) =====")
lr2 = LogisticRegression(class_weight='balanced', max_iter=1000, random_state=42)
lr2.fit(X_train2_scaled, y_train2)
yp = lr2.predict(X_test2_scaled); ypp = lr2.predict_proba(X_test2_scaled)[: ,1]
print(f"Accuracy: {accuracy_score(y_test2,yp):.3f} F1: {f1_score(y_test2,yp):.3f} ROC-AUC: {roc_auc_s

# --- Random Forest ---
print("\n==== RANDOM FOREST (full features) =====")
rf2 = RandomForestClassifier(n_estimators=200, class_weight='balanced', random_state=42, n_jobs=-1)
rf2.fit(X_train2, y_train2)
yp = rf2.predict(X_test2); ypp = rf2.predict_proba(X_test2)[: ,1]
print(f"Accuracy: {accuracy_score(y_test2,yp):.3f} F1: {f1_score(y_test2,yp):.3f} ROC-AUC: {roc_auc_s

==== LOGISTIC REGRESSION (full features) =====
Accuracy: 0.681 F1: 0.522 ROC-AUC: 0.730

==== RANDOM FOREST (full features) =====
Accuracy: 0.732 F1: 0.423 ROC-AUC: 0.709
```

✓ Step 11: Third Model (XGBoost) and Hyperparameter Tuning

We add a third algorithm, XGBoost, which often performs best on tabular data. We also tune its key hyperparameters using cross-validation to find the best configuration, as required by the assignment.

```
# =====
# STEP 11: XGBoost + Hyperparameter Tuning
# =====
# Install xgboost if needed
!pip install xgboost --quiet

from xgboost import XGBClassifier
from sklearn.model_selection import GridSearchCV

# 11.1 Handle class imbalance with scale_pos_weight
# ratio = negatives / positives
ratio = (y_train2 == 0).sum() / (y_train2 == 1).sum()
print(f"scale_pos_weight (imbalance ratio): {ratio:.2f}")

# 11.2 Define XGBoost with a small hyperparameter grid
xgb = XGBClassifier(
    scale_pos_weight=ratio,
    eval_metric='logloss',
    random_state=42,
    n_jobs=-1
)

param_grid = {
    'n_estimators': [100, 200],
    'max_depth': [4, 6],
    'learning_rate': [0.1, 0.05]
}

# 11.3 Grid search with 3-fold cross-validation, optimising ROC-AUC
print("Running grid search (this may take 1-2 minutes)...")
grid = GridSearchCV(xgb, param_grid, cv=3, scoring='roc_auc', n_jobs=-1)
grid.fit(X_train2, y_train2)

print(f"\nBest parameters: {grid.best_params_}")
print(f"Best CV ROC-AUC: {grid.best_score_:.3f}")

# 11.4 Evaluate the best model on the test set
best_xgb = grid.best_estimator_
yp = best_xgb.predict(X_test2)
ypp = best_xgb.predict_proba(X_test2)[: , 1]
print("\n==== XGB00ST (tuned) on test set =====")
print(f"Accuracy: {accuracy_score(y_test2, yp):.3f}")
print(f"Precision: {precision_score(y_test2, yp):.3f}")
print(f"Recall: {recall_score(y_test2, yp):.3f}")
```

```

print(f"F1-Score: {f1_score(y_test2, yp):.3f}")
print(f"ROC-AUC: {roc_auc_score(y_test2, ypp):.3f}")

scale_pos_weight (imbalance ratio): 2.64
Running grid search (this may take 1-2 minutes)...

Best parameters: {'learning_rate': 0.1, 'max_depth': 4, 'n_estimators': 200}
Best CV ROC-AUC: 0.750

===== XGB00ST (tuned) on test set =====
Accuracy: 0.694
Precision: 0.462
Recall: 0.676
F1-Score: 0.549
ROC-AUC: 0.758

```

✓ Step 12: Model Evaluation Visualisations

We visualise the best model (XGBoost) with a confusion matrix, ROC curve, and feature importance chart. These show how well the model performs and which features are most predictive of game success.

```

# =====
# STEP 12: Model Evaluation Visualisations
# =====

import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay, roc_curve

# Use the best model (tuned XGBoost)
y_pred = best_xgb.predict(X_test2)
y_proba = best_xgb.predict_proba(X_test2)[: , 1]

fig, axes = plt.subplots(1, 3, figsize=(18, 5))

# --- 1. Confusion Matrix ---
cm = confusion_matrix(y_test2, y_pred)
ConfusionMatrixDisplay(cm, display_labels=['Not Success', 'Success']).plot(ax=axes[0], cmap='Blues')
axes[0].set_title('Confusion Matrix (XGBoost)')

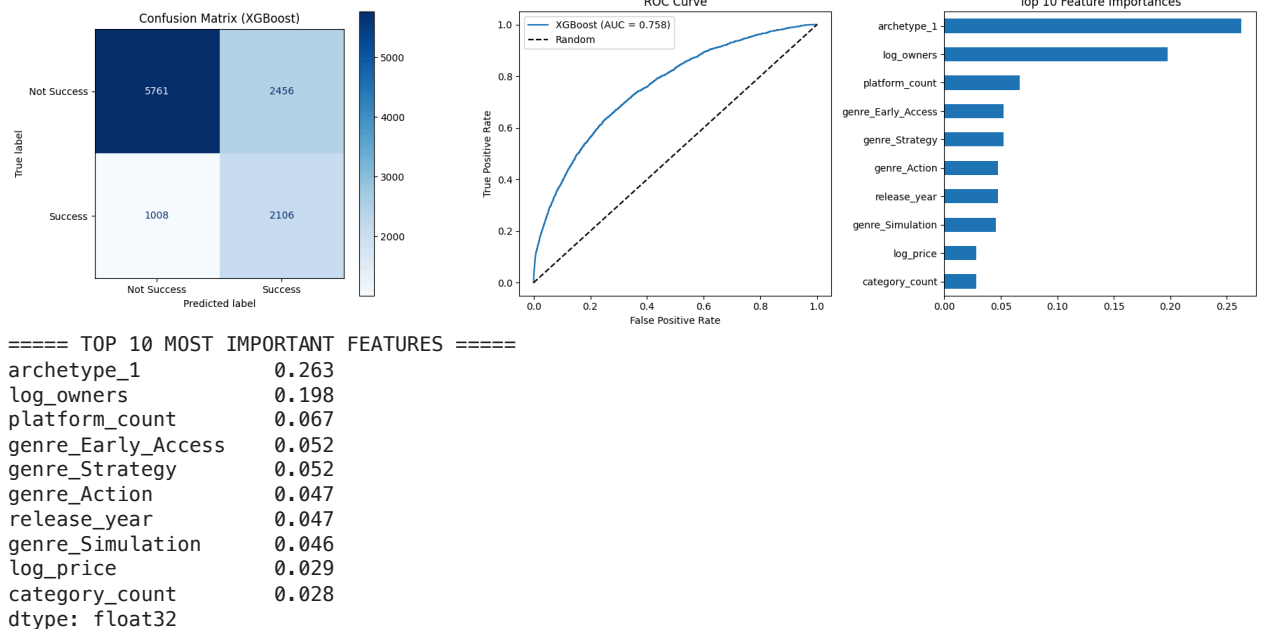
# --- 2. ROC Curve ---
fpr, tpr, _ = roc_curve(y_test2, y_proba)
axes[1].plot(fpr, tpr, label=f'XGBoost (AUC = {roc_auc_score(y_test2, y_proba):.3f})')
axes[1].plot([0, 1], [0, 1], 'k--', label='Random')
axes[1].set_xlabel('False Positive Rate')
axes[1].set_ylabel('True Positive Rate')
axes[1].set_title('ROC Curve')
axes[1].legend()

# --- 3. Feature Importance (top 10) ---
importances = pd.Series(best_xgb.feature_importances_, index=X_full.columns)
top10 = importances.sort_values(ascending=False).head(10)
top10.sort_values().plot(kind='barh', ax=axes[2])
axes[2].set_title('Top 10 Feature Importances')

plt.tight_layout()
plt.show()

# Print the top features as text too
print("===== TOP 10 MOST IMPORTANT FEATURES =====")
print(top10.round(3))

```



Step 13: Saving the Clean Dataset for Tableau (Assignment 2)

We export the cleaned dataset, including the engineered features and K-Means archetype labels, to a CSV file. This file will be the input for the Tableau dashboard in Assignment 2.

```
# =====
# STEP 13 (FINAL FIX): Clean game names and export for Tableau
# =====

from google.colab import files

# 13.1 The real game name is in the 'AppID' column for 56,632 rows.
# Only 23 rows are broken (AppID is a number). We remove those.
appid_is_number = df_model['AppID'].astype(str).str.match(r'^\d+$')
df_clean = df_model[~appid_is_number].copy()
print(f"Rows after removing 23 broken rows: {len(df_clean):,}")

# 13.2 Set the correct game name
df_clean['game_name'] = df_clean['AppID']

# 13.3 Archetype names - matched to THIS run's cluster profiles
# 0 = high price + newest      -> New Premium Releases
# 1 = cheapest + lowest success -> Budget / Casual
# 2 = most multi-platform    -> Cross-Platform Games
# 3 = highest owners + oldest -> Proven Classics
archetype_names = {
    0: 'New Premium Releases',
    1: 'Budget / Casual',
    2: 'Cross-Platform Games',
    3: 'Proven Classics'
}
df_clean['archetype_name'] = df_clean['archetype'].map(archetype_names)

# 13.4 Columns to export for the Tableau dashboard (Assignment 2)
export_cols = [
    'game_name', 'Genres', 'Tags',
    'price_real', 'owners_avg', 'release_year',
    'platform_count', 'genre_count', 'category_count', 'is_free',
    'Positive', 'Negative', 'total_reviews',
    'positive_ratio', 'wilson_score',
    'success', 'archetype', 'archetype_name'
]

# 13.5 Save locally (Drive is not mounted in this notebook) and download
output_path = '/content/steam_clean.csv'
df_clean[export_cols].to_csv(output_path, index=False)
print(f"Saved: {output_path} | Rows: {len(df_clean):,}")

# Trigger browser download
```

```
files.download(output_path)
```

```
print("\nPreview:")
df_clean[export_cols].head()
```

Rows after removing 23 broken rows: 56,632
Saved: /content/steam_clean.csv | Rows: 56,632

Preview:

	game_name	Genres	Tags	price_real	owners_avg	releas
496350	Supipara - Chapter 1 Spring Has Come!	Adventure	Adventure, Visual Novel, Anime, Cute	5.24	10000.0	
	Mvsterv					